



UN Climate Change report

The call from UN for all countries to come together and take drastic and immediate action to limit emissions. <https://digit.in/aug20-13>



This could happen again

Interview with Madhav Gadgil on the mismanagement of dams during the Kerala Floods. <https://digit.in/aug20-14>

In a 2019 “emissions gap” report, the UN called for drastic global action to limit a rise in temperatures beyond 1.5 degrees Celsius as compared to pre industrial times. This is because despite international agreements, real action on reducing emissions and curtailing global warming has been lacking. While climate change affects everyone on the planet, there are some regions where the effects are more drastic. As a consequence, the changes in these regions are better documented, and are known as climate change hotspots.

AROUND THE WORLD

With climate change, the existing patterns in weather are exaggerated. What happens is that the regions that receive a lot of rainfall, get even more rainfall, and the regions that are hot and dry, tend to be baked even further. All around the world, this causes problems such as draughts, forest fires, floods and habitat loss. The periods of draughts tend to get longer, during

which time the precipitation of water vapour in the atmosphere increases inordinately, followed by extreme weather events such as cyclones and snowstorms. This is why Mumbai saw the first cyclone in over a hundred years, earlier in the year. The dry areas continue to become drier as there is increasingly less water available for precipitation from these areas. The regions of the world below the tropics that are dry and arid, are becoming increasingly so. Africa and Asia are the worst affected, with the desert regions here expanding, along with an increase in draughts. Environmental models suggest that these extreme weather events will only increase in frequency if there are not drastic and immediate

steps taken to reduce carbon emissions. Then there are the regions close to the poles, and we know that the polar regions are among the most vulnerable to climate change. The ice caps here continue to melt, with massive regions of the arctic ice shelf chipping away. Then there is the permafrost, in Alaska and Siberia, which are also melting and disappearing. This not only causes floods, but also releases carbon dioxide and methane trapped within the ice, which further aggravates the climate change, causing even more permafrost to melt, in a feedback loop. This section in the February 2018 issue of Digit has more information on how climate change affects the permafrost. The melting glaciers and ice increase

CLIMATE CHANGE HOTSPOTS

There are some parts of the world that will be particularly affected by global warming

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Credit: StatOil

The Sleipner A project that injects carbon into a saltwater reservoir beneath the ocean